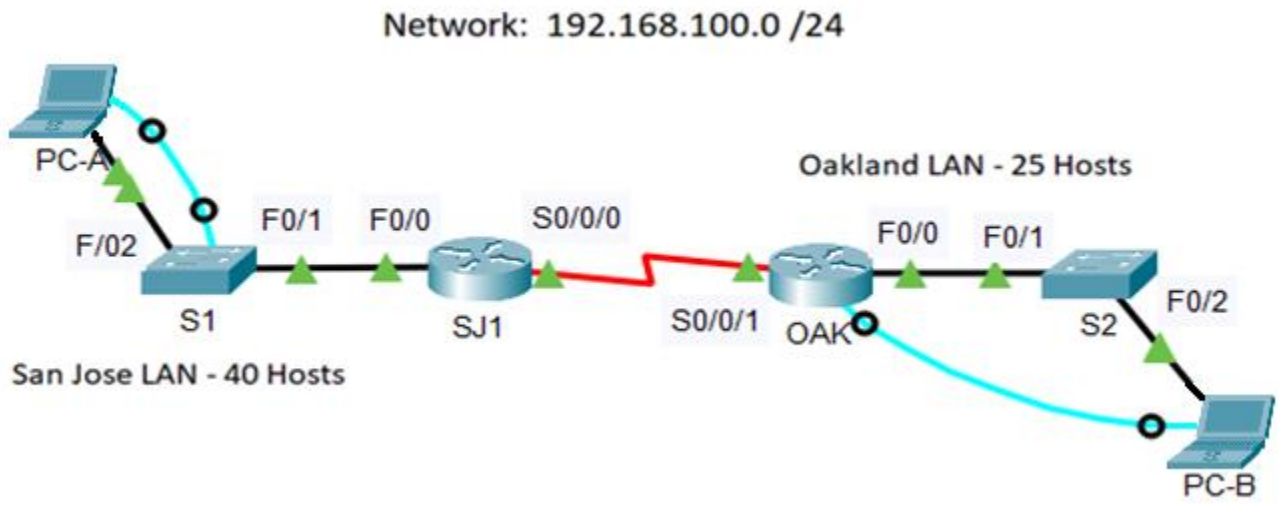
Lab - Build a LAN/WAN Network

Topology



Addressing Table

Device	Interface	IP Address / Prefix	Default Gateway
SJ1	F0/0	**192.168.100.1 /26**	**N/A**
	S0/0/0	**192.168.100.97 /30**	**N/A**
OAK	F0/0	**192.168.100.65 /27**	**N/A**
	S0/0/1	**192.168.100.98 /30**	**N/A**
S1	F0/1	**192.168.100.2 /26**	**192.168.100.1**
S2	F0/1	**192.168.100.66 /27**	**192.168.100.65**
PC-A	NIC	**192.168.100.62 /26**	**192.168.100.1**
PC-B	NIC	**192.168.100.94 /27**	**192.168.100.65**

Objectives

Part 1: Set Up the Topology and Initialize Devices

Part 2: Configure Devices and Verify Connectivity

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Background / Scenario

This is a comprehensive lab to review previously covered IOS commands. In this lab, you will cable the equipment as shown in the topology diagram. You will then configure the devices to match the addressing table. After the configurations have been saved, you will verify your configurations by testing for network connectivity.

After the devices have been configured and network connectivity has been verified, you will use IOS commands to retrieve information from the devices to answer questions about your network equipment.

This lab provides minimal assistance with the actual commands necessary to configure the router. Test your knowledge by trying to configure the devices without referring to the content or previous activities.

Required Resources

- 2 Routers (Cisco 1841)
- 2 Switch (Cisco 2960)
- 2 PCs (Lab Laptop with Putty or Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology
- Serial Cable for the WAN (details to come)

Instructions

Part 1: Set Up Topology and Initialize Devices

Step 1: Examine Network Requirements.

a. Develop a VLSM address scheme for the network displayed in the topology diagram

b. Use the 192.168.100.0/24 network address. Record your addresses using the table on the first page

c. 1st host for the default gateway (DGW), 2nd host for VLAN1s, last Host for the PCs. WAN use the 1st host for SJ.

Step 2: Cable the network as shown in the topology.

a. Attach the devices shown in the topology diagram, and cable, as necessary.

b. Power on all the devices in the topology.

Step 3: Initialize and reload the routers and switches.

If configuration files were previously saved on the routers or switches, initialize and reload these devices back to their default configurations.

Part 2: Configure Devices and Verify Connectivity

In Part 2, you will set up the network topology and configure basic settings, such as the interface IP addresses, device access, and passwords. Refer to the **Error! Reference source not found.** and **Error! Reference source not found.** at the beginning of this lab for device names and address information.

Step 1: Assign static IP information to the Windows 10 VM interfaces.

a. Configure the IP address, subnet mask, and default gateway settings on PC-A.

b. Configure the IP address, subnet mask, and default gateway settings on PC-B.

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Step 2: **Configure the routers** (using the lab PC Putty application.)

a. Console into the routers and enable privileged EXEC mode.

b. Enter configuration mode.

c. Assign a device name to the router.

d. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.

e. Assign **class** as the privileged EXEC encrypted password.

f. Assign **cisco** as the console password and enable login.

g. Assign **cisco** as the VTY password and enable login.

h. Encrypt the plain text passwords.

i. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.

j. Configure and activate the Ethernet interface on the router. (hint: F0/0)

k. Configure the WAN Serial Interfaces between the two network routers:

```
SJ1(config)#int s0/0/0
SJ1(config-if)#ip address A.B.C.D A.B.C.D
SJ1(config-if)#clock rate 128000
SJ1(config-if)#no shut

OAK(config)#int s0/0/1
OAK(config-if)#ip address A.B.C.D A.B.C.D
OAK(config-if)#no shut
```

l. Configure an interface description for each interface indicating which device is connected to it.

m. Configure the RIP version 2 routing protocol on each router (more later)

```
SJ1(config)#router rip
SJ1(config-router)#network 192.168.100.0
SJ1(config-router)#version 2
SJ1(config-router)#exit

OAK(config)#router rip
OAK(config-router)#network 192.168.100.0
OAK(config-router)#version 2
OAK(config-router)#exit
```

n. Ping PC-B from a command prompt window on PC-A. (hint: on the Win10 virtual machine)

Were the pings successful? Explain.

The pings were not successful because the switches are not configured yet and therefore the ping does not make it past the switch to the router.

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Step 3: Configure the switches (using one of the lab PCs).

In this step, you will configure the hostnames, the VLAN 1 interfaces and default gateways.

a. Console into each switch and enable privileged EXEC mode.

b. Enter configuration mode.

c. Assign a device name to each of the switches.

d. Disable DNS lookup to prevent the switch from attempting to translate incorrectly entered commands as though they were host names.

e. Configure and activate the VLAN interfaces on the switches.

f. Configure the default gateways for the switches

g. Save the running configurations to the startup configuration files.

Step 4: Verify connectivity end-to-end connectivity.

a. From PC-A, ping PC-B.

b. From S1, ping PC-B.

All the pings should be successful.

Part 3: Display Device Information

In Part 3, you will use **show** commands to retrieve interface and routing information from the router and switch.

Step 1: Display the routing table on each router.

a. Use the **show ip route** command on each router to answer the following questions.

What code is used in the routing table to indicate a directly connected network?

The code is C

What code is used in the routing table to indicate other networks?

There are two on each router. On SJ1 they are 192.168.100.0/26 and 192.168.100.96/30. On the OAK router they are 192.168.100.64/27 and 192.168.100.96/30

How many route entries are coded with a C code in the routing table?

A Fastethernet interface and a serial interface for the LAN side and the WAN side

What interface types are associated with the C coded routes?

1 on each router. SJ1 learned 192.168.100.64/27 and OAK learned 192.168.100.0/26

How many routes are coded with an R code on the routing table?

There is one coded R for each router. 192.168.100.64 /27 for SJ1 and 192.168.100.0 /26 for SJ1

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